



The Falcons

ORBIT — Real-time AI Incident Commander

Operational Response & Briefing Intelligence Team

PS41 | EchoSphere Hackathon 2026

Fuzail Ahmad • Lakshya Garg • Aryan Rai

**Echo
Sphere**

Team Details:

S.No	Team Member Name	Team Member Role	Team Member Mail ID
1	Fuzail Ahmad	Product & AI Orchestration	fuzailahmad2006@gmail.com
2	Lakshya Garg	Real-time Voice & Backend	lgarg7112@gmail.com
3	Aryan Rai	Frontend & Integrations	aryanbrijrajrai@gmail.com



Track - Problem Statement:

Track Chosen: Voice AI

PS41 — Real-time AI Incident Commander

An AI teammate for live operational and technical incident rooms — built to maintain shared truth, ownership, and momentum during high-pressure outages.

Agora Conversational AI = core real-time voice layer

Echo
Sphere

Problem Description:

During an outage, teams speak in fragments: facts mix with guesses, owners are unclear, decisions vanish, and updates arrive late. This slows recovery and erodes trust. Existing incident rooms capture conversation, not a defensible shared operating picture — especially when evidence conflicts and critical actions require human approval.

[50-word problem description](#)

Proposed Solution / Idea Overview:

ORBIT joins an Agora-powered live incident room as a multilingual conversational AI teammate. It recognizes speakers and roles, structures every update as a confirmed fact, hypothesis, decision, or action, and maintains a live Incident Digital Twin with provenance and confidence. Specialized agents reconcile conflicts, expose what is still unknown, build the timeline, track owners, retrieve runbooks and similar incidents, and call approved tools across Slack, Jira, PagerDuty, and monitoring. The Commander speaks concise status summaries at useful moments, while human approval gates every critical action. ORBIT never claims a root cause without evidence: it organizes, correlates, recommends, and keeps the team aligned until recovery and post-incident review.

[Proposed solution / idea overview \(134 words\)](#)

Technical Architecture

One live room. One trusted incident state.



Planned use of Agora: the core low-latency voice room, conversational interface, and spoken incident briefings. WebSockets and RBAC complete the production-ready layer.

Specialist agents keep the room coordinated.

Listener

speaker + role

Evidence

facts + provenance

Conflict

gaps + contradictions

Timeline

Digital Twin

Action

owners + escalation

Investigation

tools + RAG

Commander

spoken briefings

Listen → understand → structure → verify → recommend → human approves → act → monitor → learn

Traceable evidence at every step.

Shared truth

Confirmed facts vs assumptions / hypotheses
Decisions, action items, ownership
Live incident timeline

Safety & alignment

Missing-info and contradiction graph
“What We Still Don’t Know” engine
Human confirmation for critical actions

Deep intelligence

Evidence Knowledge Graph + confidence
Historical memory, vector DB, similar incidents
Runbooks, anomaly correlation, severity + blast radius

Close the loop

Action aging / escalation
Recovery verification + decision audit trail
Incident replay, analytics, automatic postmortem

Demo: payment outage becomes shared truth.

14:31	Backend: "Payment API is failing."	Confirmed fact + timeline event
14:33	DevOps: "DB pool may be exhausted."	Hypothesis — not confirmed
14:34	Metrics show normal DB saturation.	Conflict surfaced; evidence linked
14:35	"Fuzail, inspect gateway logs."	Action assigned; owner + follow-up
14:38	Commander speaks a status summary.	No root cause claimed; risks remain visible
Recovery	Human approves mitigation; system verifies.	Final summary + unresolved risks

Additional Notes: Complete Prototype Scope

ORBIT = Operational Response & Briefing Intelligence Team

Live room intelligence

Agora real-time voice • speaker + role recognition • multilingual conversations • conversational agents • spoken status summaries

Investigation & prediction

Evidence Knowledge Graph • “What We Still Don’t Know” • RAG/vector memory • similar incidents • runbooks • anomaly correlation • blast radius • severity prediction • screenshot/log/chart analysis

Evidence & coordination

Facts / hypotheses / decisions / actions • provenance • confidence • Digital Twin • timeline • owners • follow-ups • contradiction + missing-info detection

Controlled operations & learning

Jira / Slack / PagerDuty / monitoring tool calls • WebSocket dashboard • RBAC • human approval • action aging • recovery verification • audit trail • replay • analytics • automatic postmortem

Expected Impact & Future Vision: Faster alignment, safer evidence-led decisions, clearer accountability, and a living incident memory that helps every team recover and learn faster after each outage.